Zoo 404 physiology (Spring -2022)

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- **Question 01:**

What are conformers write its types ??

- **Ans:**

Living organisms which cannot maintain a constant internal environment are Known as "conformer".

The animals that conform to external changes are unable to maintain

Homeostasis for internal condition.

Types:

- **Osmoconformers**
- **Oxyconformers**
- **Thermoconformers**

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- **Question 02:**

what is the Role of ADH?

Ans:

Antidiuretic Hormone (ADH) also known as Vasopressin.

Role:

It promotes water reabsorption in the kidneys and decreases urine volume, helping to regulate blood osmolarity.

It also acts as a vasoconstrictor, increasing the arterial pressure. For this reason, it is named vasopressin.

- **Question 03:**

How does the resistance in blood flow can be overcome by pressure ?

Ans:

When heart contracts, pressure in the ventricles increases. This pressure is

used to overcome resistance to flow through the vessels.

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◦ **Question 04:**

How B12 is absorbed in small intestine?

Ans:

Vitamin B12 (molecular weight 1357) is the largest water-soluble essential

nutrient taken up intact. B12 absorbed across the intestinal lumen in distal ileum.

In the stomach, hydrochloric acid and enzymes unbind vitamin B12 into its free form. From there, vitamin B12 combines with a protein called intrinsic factor so that it can be absorbed further down in the small intestine.

◦ **Question 05:**

What is quantal released of neurotransmitters?

Ans:

Neurotransmitters are generally released in tiny packets called quanta. Each quantum may consist of about 2000 to 10,000 molecules of transmitter molecules.

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◦ **Question 06:**

Write two mechanisms of motility?

Ans:

Alimentary canal motility is achieved by two different mechanisms:

• **Muscular motility:** Involves contractions of muscles in the walls of alimentary canal. It Permits handling of harder and larger pieces of food

• **Ciliary motility:** Involves beating of cilia lining the digestive tract. Cilia generate fluid

Currents. Fluid current propels food in digestive tube.

◦ **Question 07:**

Write functions of Alimentary canal?

The alimentary canal is a muscular hollow continuous tubular organ that starts at the mouth and terminates at the anus and is responsible for the digestion and absorption of the ingested food and liquids.

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3 marks Descriptive Question

- **Question 01:**

Define donnan effect?

Donnan equilibrium implies that:

If there is a nondiffusible solute in one of the sol. separated by semi-permeable membrane, the conc. of sol. on both sides do not equalize. Concentration of solution with non-diffusible solute remains high even at equilibrium. This is Known as Donnan effect.

- **Effects of Donnan equilibrium:**
- The unequal distribution of solutes results in the movement of water into the cell, increasing its hydrostatic Pressure.
- Unequal distribution of ions results in The development of electrical potential across the plasma membrane.

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- **Question 02:**

Discuss ornithine urea cycle?

Ornithine-Urea Cycle is used by all vertebrates except teleost fishes. It Synthesize urea primarily in the liver. Two ammonia and one CO_2 molecule are added to ornithine to form Arginine. Enzyme arginase cleaves arginine to remove a molecule of urea and regenerate ornithine.

- **Question 03:**

Differentiate between endopeptidases and exopeptidases?

i. Endopeptidases:

Attack peptide bonds within protein molecule. It break large peptide chains into shorter polypeptide segments

ii. Exopeptidases:

Attack terminal peptide bonds of shorter polypeptides. It Release free amino acids or dipeptides and tripeptides

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- **Question 04:**

Write the factors of ultra filtration?

Filtration takes place entirely passively across three layered wall of the glomerulus into the capsular space.Filtration depends on three factors:

- Net hydrostatic pressure
- Colloidal osmotic pressure
- Permeability of filtration membrane

- **Question 05:**

Name constitutes of respiration in lungs?

Respiratory portion of the lung constitutes

- Terminal bronchioles
- Respiratory bronchioles
- Alveolar ducts
- Alveolar sacs

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- **Question 06:**

What are the uses of Kymograph?

"Kymograph is a device that graphically records changes in the mechanical activities of animal tissues"

- **Uses:**

Kymograph used in physiological experiments related to:

- Study of skeletal muscle contractions (twitch and tetanus)
- Cardiac muscle activities (cardiac cycle)
- Measurement of blood pressure
- Rate of respiration

- **Question 07:**

What is synapse?

Synapses are the Junctions between axon terminals of one neuron and the dendrites of another neuron. Electrical synapses and Chemical synapses are two types of synapse.

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- **Question 08:**

Define Motility and it's significance?

- **Motility:**

"Ability of the alimentary tract to contract and propel ingested material along its length". It depends on activity of smooth muscle walls and/or ciliary lining of digestive tract.

- **Significance of Motility:**

It is significant for:

- **Translocation and conduction of food**
- **Mechanical treatment of food**
- **Prevent scumming**

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5 Marks Long Question

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Long Question (5 marks)

- **Question 01:**

Write a note on lipases?

Lipases are lipid and fat digesting enzymes. Which secreted in pancreatic Juice and by intestinal epithelium. Lipases act on emulsified fat droplets and hydrolyze them into:

- Monoglycerides
- Diglycerides
- Free fatty acids and glycerol

The emulsification by bile salts is necessary for lipases to hydrolyze lipids

- If bile salts are insufficient, fats are not emulsified
- Fat digestion by lipases remains incomplete

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- **Function of lipases:**
- Lipase is an enzyme that breaks down triglycerides into free fatty acids and glycerol by catalyzing the hydrolysis of the ester bonds in triglycerides.
- Lipases are present in pancreatic secretions and participate in fat digestion and metabolism. They play an essential role in lipid transport and serve individual functions in several tissues.
- Hepatic lipase in the liver degrades the triglycerides that remain in intermediate-density lipoprotein (IDL).

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- **Question 02:**

Discuss P-wave and P-R interval?

- **P Wave:**

Caused by electrical potentials when atria depolarize. Occurs just prior to contraction of atria

Ventricles are in diastole during the expression of P wave.

It is typically a small positive deflection from the isoelectric baseline that occurs just before the QRS complex.

- **P-R Interval:**

P-R interval is the Period of time from start of P wave to beginning of the QRS complex.

It indicates amount of time required for SA depolarization to reach the ventricles.

The normal PR interval is between 120 – 200 ms (0.12-0.20s) in duration (three to five small squares). If the PR interval is > 200 ms, first degree heart block is said to be present

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- **Question 03:**

Describe the absorptive Surface area in small intestine?

Epithelial lining of intestine is specialized for absorptive functions. It has:

- Increased surface area for absorption
- Rich supply of transport system

The adaptations to increase Surface area are enhanced by anatomical features:

- Intestinal folds, Villi, Epithelial Cells of Villi, Microvilli

Intestinal folds, villi and microvilli collectively increase the surface area more than 500 times

The Total area for absorption in small intestine of human is 200-300 m^2.

- **Comparison**: gross cylindrical area of small intestine is only 0.4 m^2.

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Question 04:

Write a Note on blood vessels?

Blood vessels carry blood during circulation

There are three types:

1. **Arteries**
2. **Veins**
3. **Capillaries**

- **Arteries** carry blood away from heart to organs and tissues. Arteries are the blood vessels that bring oxygen-rich blood from your heart to all of your body's cells. They play a crucial role in distributing oxygen, nutrients and hormones throughout your body.
- **Veins** are blood vessels located throughout your body that collect oxygen-poor blood and return it to your heart. Veins are part of your circulatory system. They work together with other blood vessels and your heart to keep your blood moving. Veins hold most of the blood in your body.

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- **Capillaries** are delicate blood vessels that exist throughout your body. They transport blood, nutrients and oxygen to cells in your organs and body systems. Capillaries are the smallest blood vessels in your vascular system.

- **Question 05:**

Write a note on pH regulation?

PH regulation is The processes operating in living organisms to regulate the concentration of hydrogen ions and preserve a viable acid-base state.

Regulation of body fluid pH is one of the most important physiological functions of homeostasis, because activity of most chemical reactions via enzyme proteins is dependent on fluid pH.

pH of blood is maintained within a narrow alkaline range 7.35 to 7.45. Outside this range, pH becomes incompatible with life. Enzymes and proteins are denatured.

Kidneys maintain pH by two ways:

- Reabsorption of bicarbonate ions
- Removal of fixed acids and H^+

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